Trail Priority Tool

User & Maintenance Guide

Golden Gate National Recreation Area

This guide is for anyone using or maintaining the Trail Priority Tool — park staff browsing trail data, and future interns responsible for keeping it up to date. No coding or GIS background is assumed. Wherever something technical comes up, it's explained in plain terms, with a short glossary at the end for quick reference.

What This Tool Is

The Trail Priority Tool is a website that helps GGNRA staff figure out which trails most need attention. It brings together several different kinds of information about every trail in the park — how much it would cost to rebuild, how much deferred maintenance it has, what capital projects are already planned nearby, what wildlife has been spotted along it, what coastal hazards (flooding, erosion, tsunami risk) affect it, and more — and lets you sort, filter, and compare trails using any combination of that information.

It's built to answer questions like: "Which trails combine poor condition with high wildlife value?" or "Which trails haven't gotten any investment in years?" — questions that are hard to answer by looking at any single spreadsheet on its own.

Getting Started

Open the tool's web address in any browser. You'll be asked to sign in with your NPS/ArcGIS Online (AGOL) account — this is the same login you'd use for other Esri/ArcGIS tools. Signing in is required because some of the data (project funding amounts, specific figures) is internal and not meant to be public.

> *If sign-in fails or the page gets stuck after you log in, see the Troubleshooting section near the end of this guide before asking for help — most issues have a quick fix.*

Using the Tool

The trail list and sorting

The left side of the screen lists every trail. At the top, you'll find checkboxes for different ways to prioritize the list — replacement value, deferred maintenance, condition, climate risk, nearby project activity, native or endangered species nearby, hazard exposure, and more.

- Check one box to sort by just that measure.
- Check several boxes at once and the tool combines them into a single score — useful for finding trails that rank high on more than one concern at the same time.
- The ↑/↓ button flips the order, so you can just as easily find the trails with the least attention (fewest projects, lowest funding) as the ones with the most.

Filtering

Below the sort options, you can narrow the list to historic assets only, Trails Forever projects only, or trails affected by any combination of the coastal hazard layers. There's also a plain text search box for finding a trail by name, area, or county.

Clicking on a trail

Click any trail — from the list or directly on the map — to see everything about it in one place: its full statistics, nearby capital projects, recent wildlife sightings from iNaturalist, and (for San Francisco trails) recent 311 service requests nearby.

The hazard layers

Above the map, you can turn on any of five coastal hazard layers — FEMA flood zones, flooding potential, shoreline and cliff retreat projections, sea-level rise inundation, and tsunami hazard areas. These come from a formal vulnerability study (cited under "Data Sources" at the top of the tool) and show which trails sit inside which hazard zones.

Finding a species

The "Find a species park-wide" search lets you look up any plant or animal by name and see every place it's been spotted near a GGNRA trail, based on live data from iNaturalist. This is useful for understanding which trails pass through sensitive habitat.

Sketching a possible reroute

If you're thinking through moving a trail — say, away from an eroding cliff edge — the reroute tool lets you sketch a rough new path directly on the map and see what capital projects and wildlife activity are near that hypothetical route, before anyone commits to a real survey.

How This Tool Was Built

It helps to know where each piece of information actually comes from, since that affects how much to trust it and how it gets updated.

Trail condition & funding	From the PMIS/FMSS Asset Management Reporting System — the same system NPS uses park-wide to track facility condition and capital project funding.
Nearby projects	Also from PMIS/FMSS, matched to trails by real-world distance.
Wildlife sightings	Live from iNaturalist, a public citizen-science database — fetched fresh every time someone opens the tool, not stored.
311 complaints	Live from San Francisco's public 311 system — San Francisco trails only, since that's the only jurisdiction it covers.
Coastal hazards	From a formal vulnerability assessment conducted by Western Carolina University for GGNRA.
Full trail network	From the National Park Service's own public trail-mapping service.

The sensitive data (trail condition figures, project funding) lives in ArcGIS Online (AGOL), in a part of your organization's account that only signed-in staff can see. The tool's actual webpage is public and contains no sensitive numbers itself — it just asks AGOL for the real data after you sign in. This is why signing in is required, and why the page won't show any dollar figures or project details until you do.

Keeping the Data Current

Trail and project data doesn't update itself — someone needs to pull a fresh export from the source system and run it through the tool's update process periodically (for example, once a quarter, or whenever a major batch of projects gets funded). This section walks through exactly how, written for someone doing it for the first time.

The three-step process, in plain terms

- Step 1 — Export the latest project data from the FMSS reporting portal.
- Step 2 — Run a script that turns that export into a format the tool understands.
- Step 3 — Run a second script that pushes that data to ArcGIS Online, where the live tool reads it from.

Once step 3 finishes, the update is done — anyone who opens the tool from that point on sees the new data automatically. There's nothing else to publish or deploy.

Step 1: Export from FMSS

Log into the PFMD portal with your own NPS credentials, and navigate to AMRS → Project Management → FMIS/PMIS Regional Project. Run the report, then export it as an .xlsx (Excel) file — not the CSV button, which has repeatedly only exported the report's cover page instead of the actual data. Before moving on, open the file and scroll past the first page to confirm it actually contains real project rows.

Step 2 & 3: Run the update scripts

This uses GitHub Codespaces — a coding environment that runs entirely in your web browser, so nothing needs to be installed on your computer.

- Go to the "trail-data-sync" repository on GitHub.
- Click the green Code button → Codespaces tab → "Create codespace on main" (or reopen an existing one if you've used this before).
- Drag the .xlsx file you just exported into the file list on the left.
- Open a terminal (Terminal menu → New Terminal) and run the three commands below, one at a time.

```
pip install requests openpyxl

python3 fmss_to_json.py "the-file-name-you-uploaded.xlsx"

python3 push_to_agol.py
```

The second command reads your export and writes two files with the trail and project data in the right shape. The third command pushes that data live. You should see a summary ending in “All done” if everything worked.

> *You should never need to type a password or secret code during this process. The credential that lets the script talk to ArcGIS Online is stored once, securely, at the repository level (a “Codespaces secret”) and is applied automatically. If you're ever asked to type one in manually, or if you see one displayed on screen, stop and let whoever manages the AGOL account know — a credential like that should be treated like a password and regenerated if it's ever been exposed.*

If something goes wrong

“Very few components parsed” warning	The .xlsx you uploaded is almost certainly the cover-page-only export. Re-export from the FMSS portal and confirm it has real rows before trying again.
“Delete batch failed” / “not supported” error	Editing isn't turned on for one of the ArcGIS Online layers. In AGOL, open that layer's item → Settings → Editing, and enable Create, Update, and Delete.
A token or “invalid client” error	The stored AGOL credential is missing, wrong, or expired. This needs attention from whoever manages the AGOL account — see the Glossary entry for “Client Secret” for what this actually is.
The live tool shows “data failed to load” after signing in	Usually means one of the three ArcGIS Online data layers isn't shared with your whole organization. In AGOL, open each layer's item → Share, and confirm it's shared at the Organization level, not just with one individual.

Making Bigger Changes to the Tool

The steps above cover routine data updates — the most common thing anyone will need to do. Changing how the tool actually looks or behaves (adding a new feature, changing colors, adjusting how something is calculated) means editing the tool's underlying code, which lives in a GitHub repository as a single webpage file.

This entire tool — every feature described in this guide — was originally built using Claude, an AI assistant from Anthropic, through natural back-and-forth conversation rather than by writing code from scratch by hand. That's a genuinely practical way to continue maintaining it: describing what you want changed in plain English to an AI coding assistant, having it make the change, and checking the result, works well for a tool like this one and doesn't require you to personally know how to program.

If you go this route, it helps to mention that the tool is a single HTML file that fetches its data live from ArcGIS Online, and to share this guide for context. For anything involving money, security, or how sign-in works, it's worth having a second person review the change before it goes live, the same as you would for any change to a system handling internal data.

Quick Glossary

AGOL / ArcGIS Online	Esri's cloud mapping platform. This is where the tool's sensitive data (trail condition, project funding) actually lives, separate from the tool's own webpage.
Repository (“repo”)	A project's folder on GitHub, containing its code and/or files. This tool's website code lives in one repo; the update scripts live in another.
Codespace	A coding environment that runs in your browser, hosted by GitHub. Used for running the update scripts without installing anything locally.
API	A structured way for one piece of software to ask another for information — for example, how the tool asks iNaturalist for wildlife sightings.
Client Secret / credential	A password-like code that lets the update script prove to ArcGIS Online that it's allowed to make changes. Should never be shared, emailed, or pasted anywhere outside its secure storage location.
Layer (in AGOL)	A dataset published to ArcGIS Online — this tool uses separate layers for trails, projects, and hazard data.

Sharing / permissions	AGOL's settings controlling who can see or edit a given layer. Most tool problems after a fresh setup trace back to a layer being shared with only one person instead of the whole organization.

> *This guide reflects the tool as of its initial build. If features are added or the update process changes, this document should be updated to match — it's meant to stay current, not be a one-time snapshot.*